PIXEL BUG

User Manual
Version 1.5.34

Pixel art, animation, voxel models, print layouts, and small games.

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1. Getting started

Pixel Bug is a desktop editor for pixel art and sprites. It also includes tools for frame animation, tilemaps, 3D voxel models, printable layouts, and small games. Most work begins in **Base Mode**, where you draw on a pixel canvas. The other workspaces use that artwork or provide their own editing tools.

Opening Pixel Bug

If you have the Windows installer, run it and open **Pixel Bug** from its shortcut. The source-code package can also be run directly:

1. Install **Node.js 22.12.0 or later**.
2. Extract the source archive and open a terminal in the `PixelBugv1.5.34` folder, the folder containing `package.json`.
3. Run `npm install`.
4. Run `npm start` to open the desktop application.

Developers can create a Windows installer with `npm run build`. The installer is written to the `dist` folder. These commands are for the source package; they are not needed when using a prebuilt installer.

Your first drawing

1. Open **File** → **New**, or use the new project already on screen.
2. Under **Canvas Size**, choose a starting size such as **32 × 32** or **64 × 64**.
3. Choose **Pencil** from **Tools**. Pick a color and leave **Brush Size** at 1 px for individual pixels.
4. Draw on the center canvas. Use **Pick Color** to sample an existing color, **Fill** to fill an area, and **Eraser** to remove pixels.
5. Select **File** → **Save**, choose a location, and save your project as a `.pxbuild` file.
6. When you want a normal image, choose **Export** → **PNG**. Saving a project and exporting an image are separate actions.

Keep the `.pxbuild` file. It is the editable project. A PNG is useful for sharing or printing but does not replace a project backup.

2. The editor and its workspaces

The main window has a toolbar across the top, drawing controls on one side, the central canvas, and frames, layers, and color controls on the other side. The panels can scroll independently. Click **Rearrange** to change the order of Base Mode panels.

Area	What it is for
File	New, Open, Save, Save As, and Recent Projects.
Export	Image, animation, spritesheet, and project exports.
Base Mode	The standard pixel editor, shown when no other mode is active.
Voxel Mode	Build and animate voxel models in a 3D preview and slice editor.

Area	What it is for
Print Mode	Set print dimensions and arrange artwork on a page.
Play Mode	Build, test, and export a small HTML game.
Mod Mode	Write, test, and package custom brushes, effects, or Play Mode interface changes.
Settings	Appearance, keyboard shortcuts, drawing assistance, accessibility, and recovery tools.

Mode changes do not export files. Use the save or export action for the result you need. The project can contain material from several workspaces.

Moving around the canvas

You can show or hide the pixel grid with **Grid On/Off**. The canvas can be displayed at a higher zoom without changing the artwork's actual pixel dimensions. The **Ref** control is for a visual reference; **Story** opens the associated story/reference view. Choose **Custom Size** when the standard canvas sizes are not suitable.

The interface also has a **Command Palette**. Press **Ctrl + K** to search for actions by name instead of locating a button. On supported systems, use **Cmd** in place of Ctrl where applicable.

3. Drawing in Base Mode

The **Tools** panel contains the main drawing tools. Select a tool by clicking it or using its keyboard shortcut. The selected color, brush size, and other settings affect the next mark you make.

Tool	Use
Pencil (B)	Draw with the current color.
Eraser (E)	Remove pixels where you draw.
Magic Eraser (K)	Erase a connected area of matching painted pixels.
Fill (G)	Fill an enclosed or matching-color area.
Pick Color (I)	Sample a color from the artwork.
Clone Stamp (C)	Alt-click to set a source point, then paint sampled image content.
Line (L)	Draw straight pixel lines.
Rec (R)	Draw rectangles.
Ellipse (O)	Draw ellipses and circles.
Select (M)	Mark pixels for an edit or transform.
Move (V)	Move selected artwork or its selection boundary.
Text (T)	Place pixel text from the **Pixel Text** controls.

Useful drawing controls

- **Brush Size** changes the width of drawing strokes. For sharply defined sprite details, start with 1 px.
- **Sym** enables vertical, horizontal, both-axis, or quad symmetry. Leave it at **Off** for ordinary drawing.
- **Grid** shows pixel boundaries while you work; it is a viewing aid, not exported artwork.
- **Pixel Text** lets you enter text, choose a size and color, and place it with the Text tool.

- **Undo** and **Redo** let you back out of changes. They are also available from the keyboard.

Drawing tip: Use separate layers for the outline, main colors, and details. That makes later changes easier than putting everything on one layer.

4. Selections, layers, and canvas changes

Selecting part of an image

Choose **Select**, then set **Selection Mode** to Rectangle, Ellipse, Lasso, Polygon Lasso, Magic Wand, or By Color. The **Combine** setting controls whether the next selection replaces, adds to, subtracts from, or intersects the current selection. **Tolerance** determines how closely colors must match; 0 means an exact color match.

Once an area is selected, use the selection buttons to copy, cut, paste, delete, fill, add a stroke or border, flip, rotate, or clear the selection. **Grow** and **Shrink** change the selection boundary. **Select Opaque**, **Select Color**, and **Invert Sel** are useful when selecting artwork by its contents rather than drawing a boundary.

For exact placement, open **Transform**. Set the position, dimensions, rotation, flip, and the frame or layer scope as needed, then choose **Apply Transform**. The transform uses nearest-neighbor scaling, which keeps pixel edges crisp. **Keep Original** makes a transformed copy rather than replacing the original selection.

With **Move** active, use the arrow keys to nudge selected pixels. **Shift** moves in larger steps. **Alt** moves only the selection boundary; **Ctrl-drag** makes a duplicate.

Managing layers

Open the **Layers** panel to add, duplicate, delete, reorder, or group layers. Use **New Folder** to group related layers, and **Move To** to place a layer in a folder. Each layer can have its own visibility, opacity, blend mode, and color adjustments.

Use **Alpha Lock** to restrict painting to existing nontransparent pixels. **Clip Below** uses the layer below as a clipping boundary. **Merge Down** combines a layer with the one below; **Flatten** combines the artwork into fewer layers. Save before merging or flattening if you may need to edit those layers separately later.

Layer color controls include hue, saturation, lightness, brightness, contrast, and gamma. **Adjust Color**, **Desaturate**, and **Invert Colors** apply edits to layer content. Blend modes include Multiply, Screen, Overlay, Difference, and several others.

Changing the canvas

Use **Canvas Modifications** for changes to the whole image:

- **Crop Selection** keeps the selected part of the canvas.
- **Trim Content** removes unused outer space around the artwork.
- **Resize Space** changes canvas dimensions and uses the selected anchor to place existing art.
- **Nearest Scale** enlarges or reduces artwork without softening the pixels.
- **Flip H**, **Flip V**, and **Rotate 90** transform the full canvas.

Resize Space and **Nearest Scale** are different: one changes the available canvas area; the other scales the artwork itself.

5. Frames and animation

Pixel Bug stores animation in frames. A frame can have multiple layers, and the timeline provides a preview of the sequence.

Make a short animation

1. Finish the first pose in Base Mode.
2. In **Frames**, choose **Copy** to duplicate the current frame, or **Add** for a new blank frame.
3. Edit the next frame. Turn **Onion** on to see nearby frames as tinted guides.
4. Set **Frame Duration** in milliseconds. For example, 120 ms displays one frame for just over a tenth of a second.
5. Add or copy more frames, then preview the animation.
6. Export as **GIF** or **APNG** when you want an animated file.

Use **Select All**, **Copy Selected**, **Delete Selected**, and **Reverse Selected** to work with groups of frames. **Apply Time** gives the selected range a common duration. **Frame Tag** can label frames for actions such as idle or attack.

Animation clips and linked cels

Animation Clip names a frame range and gives it a playback rule: **Loop**, **Once**, or **Ping Pong**. Set the start and end frames and choose **Save Clip**. A project can contain different named animation clips without needing separate files.

Link Previous makes a cel refer to the preceding artwork instead of keeping an independent version. Use **Unlink Cel** before making edits that should affect only the current cel. Onion-skin controls let you change how many adjacent frames appear, their tint colors, and their opacity.

6. Palettes, brushes, and effects

Choosing colors

The **Palette** panel holds swatches and named presets. Select a preset or create a new one with **New**. Enter a color using the color wheel or hexadecimal field; choose **Add to Preset** to keep it as a swatch. Use **Rename** and **Delete** to manage presets.

Extract collects colors from artwork. **Add Ramp** creates steps between colors. **Sort Hue** and **Sort Value** reorganize the palette. **Remap Frames** replaces one palette color with another through the selected frame range; **Swap Colors** exchanges two colors. Palettes can be imported, exported as GIMP `.gpl` files, or exported as plain text.

Brush Lab

Brush Lab provides saved brush presets and procedural patterns, including dither, noise, cluster, hatch, selective erasing, and color replacement. **Pattern Density** changes how frequently pattern marks appear; **Pattern Scale** changes the pattern-cell size. **Source Color** is used by selective erasing and replacement. Use **Save Sel** to save a brush from a selection.

Effects and cleanup

The **Effects Stack** offers **Outline**, **Shadow**, **Dither**, and **Limit Palette**. Choose an effect, select **Preview** to inspect it, then **Apply** to commit it. **Clear** removes the current effect preview.

The **Pixel Tools** section handles common cleanup operations: replace a color, make a color transparent, threshold, posterize, gradient map, palette map, and offset the current layer. Set the scope to **Active Layer**, **Visible Layers**,

Active Frame, or **All Frames** before applying a change. Where available, restrict the operation to the active selection. **Wrap Edges** is useful when moving tileable textures without leaving a gap along the edge.

7. Importing images and models

Pixelize an image

1. In Base Mode, choose **Pixelize Image**.
2. Upload an image.
3. Adjust **Pixel Grid** and **Color Levels**. Use **Fit image inside square** when appropriate.
4. Choose **Import** to bring the result into the active layer.

The pixelizer changes an imported image into a pixel-art representation. It does not replace the original file on disk.

Convert a 3D model to pixel art

Choose **3D Converter** to open **3D Model to Pixel Art**. Supply the model files, including any needed materials or textures. The converter accepts OBJ, MTL, STL, and texture images. Choose a sprite size and either model colors or a single ink color. Optional face shading gives the output more definition.

Import Six View Layers creates separate front, back, left, right, top, and isometric images. **Import Turntable View Only** creates the chosen camera view; use the X, Y, and Z rotation controls to position it before import.

Quick Voxel Preview

Voxel Preview converts Base Mode artwork into a quick 3D view. For direct 3D editing, use **Voxel Mode** instead. The two commands serve different purposes: previewing artwork and building a voxel model.

8. Tiles, maps, and terrain

Save and reuse a tile

Select the pixels you want to reuse, then choose **Tilemap → Save Sel Tile**. Choose that tile from the Tile dropdown. Turn on **Stamp** and click on the canvas to place copies. **Delete Tile** removes a saved tile from the project's tile list.

Tile Map Editor

Choose **Map Editor** to arrange saved tiles in a grid. Set map **Width** and **Height**, choose a tile from **Saved Tiles**, and paint cells on the map. Right-click a cell to erase it. You can add map layers, change their visibility, and set collision, horizontal or vertical flip, and 90-degree rotation on individual cells.

Use **Export Tiled JSON** to save a map for compatible tilemap tools; **Import Tiled JSON** opens supported map data. Saving a tile and building a tiled map are separate steps: first create the reusable tile artwork, then lay out the map.

Repeated patterns

Open **Tiles & Terrain**, then choose **Pattern**. Select a saved source tile. Choose **Fill canvas** or **Fixed grid** and set horizontal/vertical gaps, offsets, rows, columns, and straight or staggered alignment. **Show pattern on canvas** is a preview only. When the arrangement is right, choose **Apply to layer** to paint the repeat into the active layer in one undoable step.

Elevation Terrain

The **Elevation Terrain** tab turns tile artwork or a Tile Map Editor layout into a voxel landscape.

1. Choose the landscape source: **Repeat selected tile** or **Tilemap Editor layout**.
2. Select the terrain layer you want to edit: **Height**, **Biome**, or **Material**.
3. Paint heights with **Raise**, **Lower**, **Smooth**, **Erode peaks**, **Paint to weight**, or **Exact voxel height**.
4. Set base height, highest point, surface colors, terrain style, and any biome or strata options.
5. Check the preview, then choose **Create Voxel Terrain**.

Use **Undo terrain stroke**, **Redo terrain stroke**, or **Reset heights** while editing. Terrain generation is a deliberate step, not an automatic result of drawing. Existing voxels are preserved when terrain is generated. Terrain height painting is separate from the bone **Weight Paint** tool in Voxel Mode.

9. Working in Voxel Mode

Voxel Mode is a 3D editor built around an editable slice and a live model preview. Open it from the top toolbar. The workspace includes **Modeling**, **Controls**, **Selection**, **Modifiers**, **Rigging & Parts**, and **Animation**.

Start a voxel model

1. Open **Voxel Mode** and choose **Modeling**.
2. Set the model width, height, and depth in **Controls** → **Model** if you need a different volume.
3. Choose an editing **Plane** (XY, XZ, or YZ) and a **Slice**. The front canvas edits that slice; the preview shows the full 3D result.
4. Select **Paint** and choose the voxel color, material, brush size, and depth.
5. Paint voxels on the editing slice or directly in the preview using a supported tool.
6. Use the slice arrows, brackets, Page Up/Page Down, or mouse wheel to move through slices; **Previous Filled** and **Next Filled** jump to occupied slices.

An empty editing slice does not mean the model has disappeared. Choose a filled slice or use **Focus Slice**.

Modeling and view controls

Group	Controls
Modeling	Paint, Erase, Fill, Pick, Surface Paint, Extrude, Replace, Select, Box, Line, Sphere, Cylinder, Set Origin.
Camera	Front, Right, Back, Left, Top, Bottom, Iso; orbit, pan, zoom, fit, reset, focus, turntable.
Appearance	Matte, Metal, Glass, Glow, Unlit; projection, lighting, outlines, shadows, ambient occlusion, and ground plane.
Model	Grid dimensions, model origin, mirrors, filled or hollow shapes, import bounds, frame-to-slice transfer.

In the 3D preview, drag empty space or hold **Shift** to orbit; middle-drag pans and the wheel zooms. The preview's **Front**, **Right**, **Top**, and **Iso** controls are useful for inspecting depth and alignment.

Selection and modifiers

Voxel selection tools can select the whole model, connected voxels, or voxels of the same color. Copy, paste, duplicate, delete, recolor, nudge, flip, rotate, scale, or mirror the selection. **Pivot** and **Move Snap** control how transforms are applied. Save frequently reused pieces as **Stamps**.

The **Modifiers** workspace includes **Array**, **Radial**, **Noise**, and **Hollow**. A modifier uses the current selection when one exists and otherwise works on the entire model. Preview its result before choosing **Apply**.

Parts, bones, and poses

Use **Rigging & Parts** to divide a model into parts, assign voxels to parts, and set their pivots. The armature tools let you draw and edit bones, insert joints, extrude bone chains, assign parent relationships, and check a rig. **Auto Bind**, **Bind Paint**, and **Weight Paint** control which voxels respond to a bone. **Check Rig** reports problems before export.

For posing, choose a bone or part, apply movement or rotation, and save named poses in the pose library. **Reset Pose** returns the current pose controls to their base state. Poses can also be used in animation and Play Mode.

Capture 3D View to Base Frame converts the current model view into pixel artwork. Choose the intended Base Mode action and the **Pixel Art Finish** setting before capturing. This is useful when you want a posed 3D model rendered as a normal 2D sprite.

Voxel animation and export

Open **Animation** to add, duplicate, rearrange, or remove frames and edit pose motion on the timeline. Choose **Stepped** or **Smooth** interpolation, set frame times and playback behavior, and preview the sequence. Use **Import Base Frames** when you need to bring existing Base Mode frames into the voxel animation workflow.

Voxel exports include preview and transparent PNG, spin and animation GIF, animation sheet, GLB, OBJ, STL, VOX, Voxel JSON, Animation JSON, and **Blender Script**. Use **Blender Script** when you want the rigged armature workflow in Blender; use **GLB**, **OBJ**, or **STL** when another application needs geometry.

10. Making print layouts

Print Mode is for work that must fit physical dimensions: stickers, washi tape, memo pads, repeated artwork, and single prints. Set the size before exporting rather than relying on a printer to scale the image later.

Basic print setup

1. Switch to **Print Mode**.
2. Set **DPI**, **Art Width Inches**, **Bleed Inches**, and **Safe Margin Inches**.
3. Choose a **Print Preset**: Single Art, Seamless Tile, Washi Strip, Stamp Washi, Sticker Sheet, or Memo Pad.
4. Under **Print Quality**, choose **Sheet Size**, page margins, and **Art Scaling**. **Pixel Crisp** preserves blocky edges; **Smooth Print** softens enlargement.
5. Turn on the artwork, bleed/safe guides, or cutline preview as needed.
6. Open **Large Preview** to inspect the layout, then choose **Export Print PNG**.

Bleed extends artwork beyond the intended cut edge. A **safe margin** keeps important details away from the cut. Guides help check placement; they are not a substitute for checking the exported file.

Preset-specific controls

- **Washi Strip** uses width and height controls for a repeating strip.
- **Stamp Washi** adds stamp spacing, perforations, curl, edge differences, and roundness settings.

- **Sticker Sheet** uses rows, columns, and gaps to arrange copies.
- **Memo Pad** provides memo dimensions and repeat placement.
- **Seamless Tile** repeats the source artwork across the selected layout.

Templates and the large preview

Import a custom template from PNG, JPG, SVG, or a supported 8-bit PSD file. A template can serve as a preview-only guide, appear behind the artwork, or print over it. **Use template canvas size** changes the output size to the template's dimensions when selected.

In **Large Preview**, you can import Base Mode layers as separate print layers, reposition and resize them, change order, flip or rotate, adjust opacity and blend modes, and make color adjustments without modifying the original Base Mode artwork. There are separate guide and cutline preview layers. Print edits have their own Undo and Redo.

Choose **Export Cutline SVG** when you need a vector cutline file. **Export Calibration PNG** is for checking print dimensions and printer behavior before using the final artwork.

11. Building a game in Play Mode

Play Mode builds small games using artwork, placed objects, dialogue, and connected rules. The finished **Tiny Game** is one HTML file that can be opened in a web browser. Open **Play Mode** and use its workspaces: **Builder**, **Scene**, **Player & World**, **Layers**, **Objects**, **Voxel Assets**, **Characters**, **Dialogue**, and **Rules**.

A first playable scene

1. Draw an idle player sprite in Base Mode. Make walk and jump frames if you want different actions.
2. Open **Play Mode → Player & World**. Choose the idle, walk, and jump frames, then set scene dimensions, ground position, movement speed, jump strength, and gravity.
3. Open **Scene** to name the current scene or add another scene.
4. Open **Objects** and choose a sprite frame or saved voxel asset. Set the object's position, scale, and collision, then select **Place Object**. You can also click the scene canvas to set the next placement point.
5. Open **Rules** and add a starting event such as **When Player Touches**. Connect it to an action such as **Show Text** or **Go to Scene**.
6. Select **Run Game** and test the interaction. Use **Validate Game** to check for missing or inconsistent connections.
7. Choose **Export Tiny Game** to save the standalone HTML file.

The **Builder** includes a four-part checklist: World, Objects, Rules, and Test. It also reports checks that need attention. Use it if the preview does not behave as expected.

Player controls and testing

During testing, use **A/D** or the **left/right arrow keys** to move and **Space** to jump. Use the **Interact** control for character and object interactions. Supported gamepads use the left stick or D-pad to move, the primary button to jump, and the secondary button to interact.

The testing toolbar includes **Run Game**, **Reset**, **Pause**, **Step Frame**, and **Center Actor**. The **Runtime Inspector** displays the active scene, player coordinates, checkpoint, recent rule trigger, values, and inventory. Changes to rules or objects should be tested again before export.

Scenes, objects, and dialogue

A project can have multiple scenes. Scene objects may be solid, may lead to another scene, may start dialogue, or may be saved as reusable objects. **Layers** controls parallax backgrounds and other visual layers; each can have its own scale, opacity, position, and repeat setting. **Voxel Assets** brings in saved models, model parts, poses, and backgrounds from Voxel Mode.

Characters sets up non-player characters with an assigned sprite, position, facing, and dialogue entry. **Dialogue** stores reusable text events with a display name, portrait, text, and the next line to show. Use **Preview Text** to check the result before attaching a dialogue or message to a rule.

Rules: events, actions, and conditions

The **Rules** workspace opens the visual Rule Editor. A rule chain begins with an event and continues through actions or checks. For example:

When Player Touches an object → Show Text → Give Item.

Available events include scene start, object touch, and character interaction. Actions can show text, start dialogue, give or remove items, set saved values, change numbers, move the player, play audio, save a checkpoint, change scenes, or finish the game. Conditions such as **If Value Matches** and **If Player Has Item** have separate **Then** and **Else** paths.

Rules can apply to a particular scene. Choose the correct scene and object or character when creating triggers. Use the editor's validation checks and the live tester to find unconnected rules or missing references.

12. Audio Studio

In Play Mode, open **Audio Studio** to manage sound effects and music. Its tabs are **Library**, **Waveform**, **Mixer**, and **Game Cues**.

1. Use **Import Audio** to add an existing sound, or **Create Tone** to generate a sound with a sine, square, triangle, sawtooth, or noise waveform.
2. Open **Waveform** to trim a clip, change gain, apply fades, reverse, normalize, or adjust playback stretch. Choose **Save Edited Copy** to keep the edited version as another asset.
3. In **Mixer**, set master, music, and effects levels or mute a category during testing.
4. In **Game Cues**, assign background music to a scene, or choose a triggering event and select **Create Cue Rule**.
5. Open **Rules** if you want more control over when a sound starts or stops.

Game Cues supports scene start, object touch, and character interaction events. Looping audio can be stopped with a **Stop Audio** rule. Audio Studio can also export generated audio as WAV files.

13. Mod Mode

Mod Mode creates custom brushes, effects, and Play Mode interface settings. A `.pbmod` package can contain one or more of those components.

Create and test a mod

1. Open **Mod Mode** and choose **New Draft**.
2. Enter a mod **Name** and **ID**. Choose which parts to include: **Brush**, **Effect**, and/or **Play UI**.

3. Use the provided code or Play UI settings editor. **Dot Example**, **Plus Example**, **Invert Example**, and **Outline Example** can serve as starting points.
4. Choose the appropriate permissions: reading the current canvas, returning changed pixels, or changing the Play Mode interface.
5. Select **Validate Mod** to check the draft and its required permissions.
6. Use **Test Canvas**, **Stamp Center**, **Live Effect**, or **Apply Effect Once** to inspect the result. Test-canvas changes can be undone, redone, reset, or cleared.
7. Choose **Install Draft** to use the draft locally, or **Export .pbmod** to save a package. Use **Import .pbmod** to open a saved package.

The mod test code runs in a separate restricted process. Still, check the contents and permissions of any mod you import. Mod Mode automatically saves its draft separately from normal project saving.

14. Exporting and sharing work

Open **Export** from the top toolbar. Choose a file type for one output, or configure a named **Export Profile** and choose **Export Batch** when you need several formats at once.

Output	Typical use
PNG	Still sprite or current-frame image.
Spritesheet	Multiple animation frames arranged on one image.
GIF	Looping animated preview with GIF's limited color handling.
APNG	Animated PNG, useful when PNG-style transparency is needed.
WebP	Compact still-image output.
Project	Editable .pxbuild copy.
Print PNG / Cutline SVG	Print production and cutting workflows.
Tiny Game HTML	Standalone Play Mode game.
Voxel formats	Geometry, animation, and 3D interchange.

Spritesheets and atlases

Under **Spritesheet Options**, set columns, padding, margin, scale, frame names, and whether to trim frames. **Atlas Format** offers Pixel Bug or Aseprite-style JSON. Turn on **Atlas JSON** when the consuming tool needs frame coordinates as well as the image.

For existing sheets, use **Import Sheet** to slice a regular grid into frames. Use **Import Atlas** when you have the image and its matching JSON metadata. The export dialog includes an **Export Preflight** step to catch issues before files are written.

Batch profiles

Set the output name and check the formats you need, then choose **Save Profile**. Later, select that profile and choose **Export Batch**. Batch export creates the selected outputs together. Keep a `.pxbuild` project file even if your profile also produces image or animation files.

15. Saving, backups, and recovery

Project files and tabs

Use **File** → **Save** (Ctrl + S) for the current project, or **Save As** (Ctrl + Shift + S) to save a separate project file. Pixel Bug's native project extension is `.pxbuild`. **File** → **Open** opens a saved project; **Recent** lists previously opened project files. Multiple project tabs can be open at once. A tab indicates when it has unsaved changes.

Gallery in the **Project** section opens the local project gallery. **Snapshots** opens local restore points. The **Settings** window also includes rotating **Recovery Backups** and a **Create Snapshot** action.

What autosave does

Pixel Bug stores session and recovery information locally to help restore interrupted work. This is not the same as saving a separate project file. Recovery storage may be cleared when app data is removed or the program is uninstalled, and very large sessions may exceed recovery limits.

For work you care about, save a `.pxbuild` file outside the application data folder and keep another copy in your normal backup location. Do this before large merges, format conversions, or changes that would be difficult to undo.

When opening an old or unusual project

Read any warning shown during loading. A project may be opened with limitations or in read-only form if the current editor cannot safely write it. Do not overwrite the only copy of an older project until you have confirmed that its layers, frames, and other data look right.

16. Settings, touch controls, and accessibility

Choose **Settings** to change interface colors, text size, font, spacing, and control presentation. Available themes include system, original light, original dark, and custom palettes. You can save custom interface palettes for later use. **Interface Scale** changes control text and labels, not the size of the artwork canvas.

Under **Drawing Assistance**, you can show a keyboard cursor crosshair, increase cursor contrast, or stabilize pointer painting. Accessibility options include reduced interface motion, stronger focus outlines, larger controls, increased contrast, reduced transparency, heavier text, wider text spacing, and screen-reader action announcements. These preferences do not change the saved artwork or exports.

Touch Toggle

Use **Touch Toggle** in Base Mode for a compact touch layout. Draw with one finger and pan the canvas with two fingers. **Left Handed** moves the touch layout for left-handed use. Select **Base Mode** to leave the compact touch layout. Panel rearrangement remains available in touch mode.

Custom shortcuts

In **Settings** → **Keyboard Shortcuts**, click a shortcut box and press the new key combination. Pixel Bug rejects reserved application and system shortcuts. Choose **Reset Shortcuts** to return to the defaults.

17. Keyboard shortcuts

The following are the default shortcuts. They can be changed in Settings. Ctrl refers to the Control key on Windows; some systems use Command instead.

Shortcut	Action	Shortcut	Action
B	Pencil	**E**	Eraser
K	Magic Eraser	**G**	Fill
I	Pick Color	**C**	Clone Stamp
L	Line	**R**	Rectangle
O	Ellipse	**M**	Select
V	Move	**T**	Text
Ctrl + Z	Undo	**Ctrl + Y**	Redo
Ctrl + Shift + Z	Redo	**Ctrl + A**	Select all
Ctrl + C	Copy selection	**Ctrl + X**	Cut selection
Ctrl + V	Paste selection	**Delete**	Delete selection
Escape	Unselect	**Ctrl + S**	Save project
Ctrl + Shift + S	Save As	**Ctrl + O**	Open project
Ctrl + K	Command Palette	**Ctrl + ,**	Settings

Some keys work differently when the focus is inside a text field or a different workspace. Voxel Mode also has slice navigation and camera controls, and Play Mode uses movement keys while its game tester is active.

18. Common problems and file formats

I exported an image, but I cannot edit its layers.

Open the original `.pxbuild` project. Image exports are flattened or otherwise prepared for their destination and are not substitutes for the editable project.

I cannot see the artwork on the voxel editing slice.

The model may be on another plane or slice. Choose **Previous Filled** or **Next Filled**, adjust the plane, or use **Focus Slice**. The full 3D preview can still show a model while the currently selected slice is empty.

My tiles repeat in the preview but are not drawn on the layer.

Show pattern on canvas is a preview option. Choose **Apply to layer** in the **Pattern** tab to commit the repeat.

My game starts but an interaction does nothing.

Open **Rules** and check the starting event, its scene, the selected object or character, and the connection to the next rule. Use **Validate Game** and the **Runtime Inspector**, then run the tester again.

My print is the wrong physical size.

Confirm DPI, artwork width in inches, page size, and the printer's own scaling options. Use **Export Calibration PNG** to check the output before printing a full sheet.

The program closed before I saved.

Check the local recovery options or snapshots. If a recovery exists, restore it and immediately use **File → Save As** to create a separate `.pxbuild` file. Recovery cannot be relied on as the only backup.

Quick file-format reference

Extension	Purpose
.pxbuild	Editable Pixel Bug project.
.png, .webp	Still images.
.gif, .apng	Animated image exports.
.json	Atlas data, voxel data, animation data, or other structured interchange; contents depend on the command.
.gpl, .txt	Exported color palettes.
.tmj	Tiled JSON map.
.pbmod	Pixel Bug mod package.
.html	Exported Tiny Game.
.wav	Audio export.
.svg	Print cutline.
.obj, .stl, .glb, .vox	3D model interchange formats.
.py	Blender scene script export.

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